RESEARCH ARTICLE

Comparison of the Efficacy and Safety of Myasthenia Gravis Treatments

A Bayesian Network Meta-Analysis

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Abstract

Background and Objectives

In the past decade, several novel therapies have shown promise in treating generalized myasthenia gravis (MG); however, no head-to-head prospective studies have compared their efficacy and safety. To assist clinicians in navigating this complex treatment landscape, we sought to compare treatments while accounting for differences in trial populations.

Methods

We conducted a Bayesian network meta-analysis (PROSPERO ID: CRD42023428733) of MG treatments. We searched Scopus, Embase, Web of Science, CENTRAL, and MEDLINE up to May 2, 2025, for randomized trials in MG. Two independent reviewers (N.M. and M.R.) identified prospective multiarm studies of adults (18 or older) with generalized MG that reported mean treatment difference, with uncertainty, in MG-Activities of Daily Living (MG-ADL) or Quantitative MG (QMG) score change from baseline.

Results

Of 4,823 eligible studies, 27 placebo-controlled randomized trials (placebo n = 1,086; treatment n = 1,232) were included that targeted B cells, neonatal Fc receptor (FcRn), complement activity, CD40, and interleukin-6 signaling, as well as immunoglobulin G and broad immunosuppression therapy. Compared to those treated with placebo and standard of care, QMG scores decreased by an average of −3.63 (95% credible interval −4.43 to −2.77) points more in patients treated with FcRn inhibitors, −2.59 (−3.92 to −1.28) points more in patients treated with C5 complement inhibitors (C5i), and −2.50 (−5.11 to 0.12) points more in patients treated with $CD19^+$ B-cell depletion therapy (BCDT). The MG-ADL treatment effects were −1.93 (−2.21 to −1.65) for FcRn inhibitors, −1.84 (−2.48 to −1.19) for C5i, and −1.91 (−2.97 to −0.84) for $CD19^+$ BCDT. Sensitivity analyses detected potential confounding of treatment effects by age and sex. Patients treated with FcRn inhibitors had greater odds of treatment-related adverse events (odds ratio 2.20 [1.52–3.38]) compared with the placebo group, while C5i and $CD19^+$ BCDT showed odds comparable to placebo.

Discussion

FcRn inhibitors, C5i, and $CD19^+$ BCDT showed comparable efficacy in terms of QMG and MG-ADL reduction because meaningful differences were not observed between treatment classes. Our findings also highlight how differences between trial populations may confound the interpretation of study outcomes. Prospective comparative efficacy studies are warranted.

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Glossary

AChR = acetylcholine receptor; **AE** = adverse event; **BAFF** = B-cell activating factor inhibitor; **BCDT** = B-cell depletion therapy; **C5i** = C5 complement inhibitor; **CrI** = credible interval; **FcRn** = neonatal Fc receptor; **IL-6** = interleukin-6; **IST** = immunosuppressive therapy; **IVIg** = IV immunoglobulin; **MG** = myasthenia gravis; **MG-ADL** = MG-Activities of Daily Living; **MuSK** = muscle-specific kinase; **OR** = odds ratio; **PRISMA** = Preferred Reporting Items for Systematic Reviews and Meta-Analyses; **PSRF** = potential scale reduction factor; **QMG** = Quantitative MG; **SUCRA** = Surface Under the Cumulative Ranking; **TRAE** = treatment-related adverse event.

Introduction

Myasthenia gravis (MG) is an autoimmune disorder mediated by autoantibodies against antigens at the neuromuscular junction. These autoantibodies disrupt synaptic transmission and consequently lead to muscle weakness. Approximately 80% of patients with MG have autoantibodies against the acetylcholine receptor (AChR), and approximately 5%–10% have autoantibodies against muscle-specific kinase (MuSK). Even fewer may harbor autoantibodies against low-density lipoprotein receptor–related protein 4 (LRP4) or agrin, and some may have no detectable autoantibodies (classified as seronegative).[1] Historically, acetylcholinesterase inhibitors (pyridostigmine), acute immunomodulatory therapies (IV immunoglobulin [IVIg] and plasma exchange), corticosteroids (prednisone), nonsteroidal immunosuppressive therapies (ISTs; azathioprine, mycophenolate mofetil, methotrexate, cyclosporine, and tacrolimus), and thymectomy have been used to treat MG. More recently, targeted therapies against C5 complement activation (eculizumab, ravulizumab, and zilucoplan), B cells (rituximab, belimumab, and inebiluzumab), T cells (iscalimab), neonatal Fc receptor (FcRn; efgartigimod, rozanolixizumab, nipocalimab, and batoclimab), interleukin-6 (IL-6) signaling (satralizumab and tocilizumab), and B-cell maturation antigen (chimeric antigen receptor T-cell therapies) have been developed, some of which have been approved by the Food and Drug Administration for the treatment of generalized MG.[2]

Despite the wide range of treatment options and multiple positive randomized clinical trials, no prospective studies have compared their efficacy and safety. Researchers have conducted both frequentist and Bayesian meta-analyses, but these studies have limited clinical relevance because they do not account for differences in trial designs and populations.[3-7] For example, some clinical trials for MG included only AChR+ patients, while others included a mix of AChR+, MuSK+, LRP4+, and seronegative patients.[8-14] These patient populations respond to treatment differently, and existing meta-analyses have not accounted for this.[1,3-7] Furthermore, study-wide variables can confound treatment effects, even in randomized trials that balance covariates across study arms.[15] Therefore, valid comparisons of treatment effects require a thorough analysis of the effects of confounders. There is an unmet need to assess the relative efficacy and safety of MG treatments, and this article provides such a nuanced comparison by accounting for differences in trial designs and populations.

Methods

Standard Protocol Approvals, Registrations, and Patient Consents

This study was prospectively registered in PROSPERO and reported according to Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)-Network Meta-Analysis guidelines.[16]

Screening and Inclusion/Exclusion Criteria

We performed a systematic search of Scopus, Embase, Web of Science, CENTRAL, and MEDLINE up to May 2, 2025, for randomized trials in MG (eTable 1). Search results were uploaded to Covidence to remove duplicates and for screening. Two reviewers (N.M. and M.R.) performed 2 rounds of review, independently. In the first, we screened abstracts and titles for relevance (n = 2,474 irrelevant studies excluded). In the second, we conducted a full-text review of studies that met the following inclusion criteria: (1) prospective placebo-controlled studies, (2) studies that included adults (18 or older) with generalized MG, and (3) studies that reported mean treatment difference, with uncertainty (SE, SD, or CI), in MG-Activities of Daily Living (MG-ADL) or Quantitative MG (QMG) change from baseline. In both rounds, a third reviewer (B.R.) was consulted to resolve conflicts. Overall, 27 studies met these criteria. PRISMA flowchart of the screening process is shown in eFigure 1.

Risk of Bias

We used the revised Cochrane Risk of Bias 2 tool to assess the quality of the randomized trials. Two independent reviewers (W.Z. and S.H.) assessed risk of bias, and a third reviewer (N.M.) resolved conflicts. Results are displayed in eFigures 2 and 3.

Statistical Analysis

The following data were collected from the placebo-controlled studies that met the inclusion criteria: mean and standard error of difference in QMG and MG-ADL score change from baseline between treatment and placebo arms, length of study, enrollment age, percentage of female study participants, baseline MG-ADL and QMG scores, and number of treatment-related adverse events (TRAEs). AEs of all severity levels were included; however, only those deemed related to the interventional treatment by the trial investigator were analyzed. For studies that reported only unadjusted means and standard errors of individual arms instead of treatment differences, we used the data from individual arms to calculate treatment differences and standard errors. For studies that reported both unadjusted and adjusted treatment differences, we used the adjusted treatment differences if available. For studies that did not explicitly state MG-ADL or QMG change from baseline but instead presented it in a figure, we extracted the relevant data using an online tool (graphreader).[17] This was particularly relevant for the phase 2 and phase 3 trials of efgartigimod, in which we extracted the outcome measures at week 4 from graphs.[18,19] We conducted a Bayesian network meta-analysis using the "gemtc" package in R to estimate treatment effects according to the following hierarchical structure:[20]

$$\text{Layer 1}: \mu_{\text{Observed}} \sim \mathrm{N}\left(\mu_{\text{Study}}, \sigma^2\right)$$

$$\text{Layer 2}: \mu_{\text{Study}} \sim \mathrm{N}\left(\theta_{\text{Treatment}}, \tau^2\right)$$

$$\text{Layer 3}: \theta_{\text{Treatment}}, \tau \sim \text{prior distribution}$$

The continuous outcomes (QMG and MG-ADL change from baseline treatment difference) and binary outcome (AEs) were modeled using random effects, and parameters were estimated using Markov Chain Monte Carlo. We placed an informative log-normal ($\mu = -2.13$, $\sigma^2 = 1.58^2$) prior on heterogeneity in the AE model. This value was empirically determined from a study of 14,886 meta-analyses that examined the distribution of heterogeneity in meta-analyses of placebo-controlled trials, which analyzed subjective binary outcomes (such as AEs).[21] We placed a weakly informative normal ($\mu = 0$, $\sigma^2 = 10^2$) prior on the heterogeneity in these models. Model convergence was assessed using the potential scale reduction factor (PSRF), with PSRF <1.05 indicating convergence.

The effect sizes of continuous outcomes (QMG and MG-ADL scores) are reported as mean difference and 95% credible interval (CrI), whereas the effect sizes of the binary outcome (AE) are reported as an odds ratio (OR) and 95% CrI. To further compare treatments, we performed large-scale (∼100,000) simulations in which treatment effects were sampled from their respective posterior distributions to rank treatments based on Surface Under the Cumulative Ranking (SUCRA) scores. These are interpreted as the probability that each treatment has the greatest effect on patients.

We then performed a sensitivity analysis adjusting for the following study-wide covariates: average age at enrollment, percentage of female patients, and baseline QMG or MG-ADL scores. We assumed the treatment effect to be distributed according to $\mathrm{N}(\mu + \beta x, \tau^2)$ and then estimated both μ and β. We also compared treatment efficacy using data strictly from AChR+ patients, MuSK+ patients, and phase 3 trials.

We used Covidence for study screening and R version 4.1.1 for analysis.

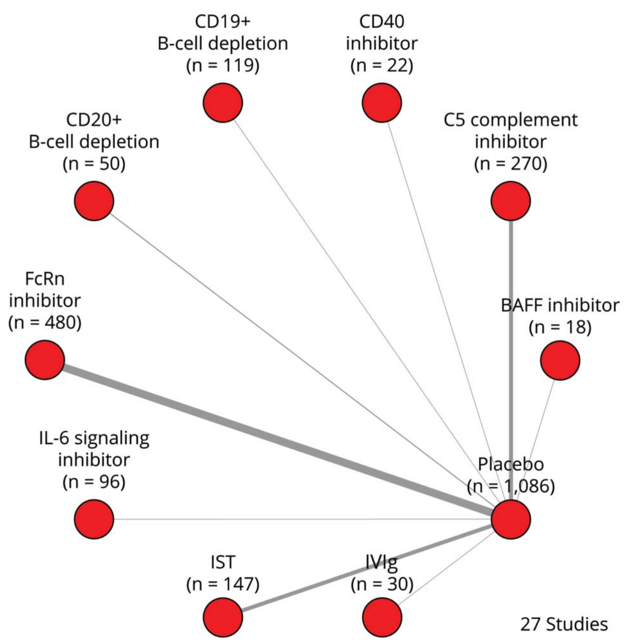
Data Availability

The data used in this study are publicly available in the articles published by the trial authors.

Results

A total of 27 placebo-controlled randomized trials (publication date 1987–2025) for 17 different treatments were included in this study (patients in placebo arms 1,086; patients in treatment arms 1,232) (Figure 1). We classified the treatments into 9 categories based on the mechanism of action: CD20$^+$ B-cell depletion (rituximab), CD19$^+$ B-cell depletion (inebilizumab), C5 complement inhibitors (C5i; eculizumab, ravulizumab, and zilucoplan), IVIg, FcRn inhibitors (nipocalimab, efgartigimod, rozanolixizumab, and batoclimab), CD40 inhibitor (iscalimab), IL-6 signaling inhibitor (satralizumab), B-cell activating factor inhibitor (BAFF; belimumab), and immunosuppressive therapy/steroid-sparing agents (mycophenolate mofetil, tacrolimus, cyclosporine, and methotrexate). Individual study data are presented in Table 1.

Figure 1 Network Plot of Included Placebo-Controlled Studies Grouped by Mechanism of Action



Edge thickness is proportional to the sample size of the treatment arm. BAFF = B-cell activating factor inhibitor; FcRn = neonatal Fc receptor; IL-6 = interleukin-6; IST = immunosuppressive therapy; IVIg = IV immunoglobulin.

Table 1 Summary Statistics of the 27 Included Placebo-Controlled Trials

Study	Phase	Mechanism of action	Treatment	Study length (wk)[a]	n	Age at enrollment, mean (SD)	Female, n (%)	Treatment-related adverse events, n	Baseline MG-ADL score, mean (SD)	Baseline QMG score, mean (SD)	Treatment effect (treatment-placebo), mean (SE): Adjusted MG-ADL score	Treatment effect (treatment-placebo), mean (SE): Adjusted QMG score
Antozzi et al., 2024[22]	2	FcRn inhibitor	Nipocalimab 5 mg/kg Q4W	8	14	NA	6 (42.9)	5	8.0 (2.8)	15.9 (2.9)	0.0 (1.1)	−0.1 (1.5)
			Nipocalimab 30 mg/kg Q4W	8	13	NA	9 (69.2)	3	8.0 (2.6)	17.1 (4.2)	−1.3 (1.1)	−0.5 (1.6)
			Nipocalimab 60 mg/kg single dose	8	13	NA	9 (69.2)	6	7.9 (2.8)	16.1 (4.1)	1.0 (1.1)	2.1 (1.6)
			Nipocalimab 60 mg/kg Q2W	8	14	NA	5 (35.7)	7	8.1 (3.2)	16.9 (2.8)	−1.3 (1.1)	−1.8 (1.5)
			Placebo	8	14	NA	8 (57.1)	1	7.3 (2.8)	17.6 (4.2)	NA	NA
Antozzi et al., 2025[13]	3	FcRn inhibitor	Nipocalimab 15 mg/kg Q2W	24	77	52.5 (15.7)	50 (65)	28	9.4 (2.7)	15.1 (4.8)	−1.5 (0.5)	−2.8 (0.7)
			Placebo	24	76	52.3 (16.4)	42 (55)	28	9.0 (2.0)	15.7 (4.9)	NA	NA
Bril et al., 2021[23]	2	FcRn inhibitor	Rozanolixizumab 7 mg/kg	4	21	50.5 (14.7)	13 (62)	10	8.2 (3.3)	16.0 (4.2)	−1.4 (0.5)	−0.7 (0.8)
			Placebo	4	22	53.3 (15.7)	14 (64)	5	6.1 (2.6)	15.4 (3.6)	NA	NA
Bril et al., 2023[14]	3	FcRn inhibitor	Rozanolixizumab 7 mg/kg	6	66	53.2 (14.7)	39 (59)	32	8.4 (3.8)	15.4 (3.7)	−2.6 (0.7)	−3.5 (1.0)
			Rozanolixizumab 10 mg/kg	6	67	51.9 (16.5)	35 (52)	39	8.1 (2.9)	15.6 (3.7)	−2.6 (0.7)	−4.8 (1.1)
			Placebo	6	67	50.4 (17.7)	47 (70)	22	8.4 (3.4)	15.8 (3.5)	NA	NA
Bril et al., 2025[24]	2	IVIg	IVIg	24	30	54.6 (17.1)	14 (46.7)	16	7.4 (3.2)	14.6 (2.8)	NA	−2.0 (1.5)
			Placebo	24	32	48 (13.7)	19 (59.3)	8	7.8 (3.4)	16.2 (4.5)	NA	NA
Gomez Mancilla et al., 2024[25]	2	CD 40 inhibitor	Iscalimab	25	22	44.7 (13.5)	12 (54.5)	11	NA	14.7 (4.2)	−1.5 (0.9)	−1.1 (1.4)
			Placebo	25	22	43.3 (13.9)	16 (72.7)	11	NA	16.3 (4.2)	NA	NA
Habib et al., 2025[26]	3	IL-6 signaling inhibitor	Satralizumab	24	96	47 (14.4)	63 (66)	45	8.1 (2.9)	15.2 (5.0)	−1.0 (0.4)	−1.6 (0.6)
			Placebo	24	92	45.9 (16.3)	56 (61)	20	7.9 (3.0)	13.6 (5.4)	NA	NA

Continued

Table 1 Summary Statistics of the 27 Included Placebo-Controlled Trials *(continued)*

Study	Phase	Mechanism of action	Treatment	Study length (wk)[a]	n	Age at enrollment, mean (SD)	Female, n (%)	Treatment-related adverse events, n	Baseline MG-ADL score, mean (SD)	Baseline QMG score, mean (SD)	Treatment effect (treatment-placebo), mean (SE)	
											Adjusted MG-ADL score	Adjusted QMG score
Hewett et al., 2018[27]	2	BAFF inhibitor	Belimumab	36	18	52.7 (17.3)	10 (56)	5	NA	NA	−0.3 (0.8)	−2.3 (2.1)
			Placebo	36	21	59 (13.9)	14 (67)	7	NA	NA	NA	NA
Howard et al., 2013[28]	2	C5 complement inhibitor	Eculizumab	16	7	NA	NA	7	NA	NA	−3.6 (1.7)[b]	−4.7 (2.8)[b]
			Placebo	16	7	NA	NA	6	NA	NA	NA	NA
Howard et al., 2017[10]	3	C5 complement inhibitor	Eculizumab	26	62	47.5 (15.7)	41 (66)	NA	10.5 (3.1)	17.3 (5.1)	−1.4 (0.7)	−2.5 (0.8)
			Placebo	26	63	46.9 (18)	41 (65)	NA	9.9 (2.6)	16.9 (5.6)	NA	NA
Howard et al., 2019[18]	2	FcRn inhibitor	Efgartigimod	4	12	55.3 (13.6)	7 (58.3)	NA	8.0 (3.0)	14.5 (6.3)	−1.7 (1.1)[b]	−3.5 (1.8)[b]
			Placebo	4	12	43.5 (19.3)	8 (66.7)	NA	8.0 (2.2)	11.8 (5.4)	NA	NA
Howard et al., 2020[29]	2	C5 complement inhibitor	Zilucoplan 0.1 mg/kg	12	15	45.5 (15.7)	8 (53.3)	8	6.9 (3.3)	18.7 (4.0)	−2.2 (1.3)[b]	−2.3 (1.7)[b]
			Zilucoplan 0.3 mg/kg	12	14	54.6 (15.5)	4 (28.6)	3	7.6 (2.6)	19.1 (5.1)	−2.3 (1.3)[b]	−2.8 (1.7)[b]
			Placebo	12	15	48.4 (15.7)	11 (73.3)	3	8.8 (3.6)	18.7 (4.0)	NA	NA
Howard et al., 2021[19]	3	FcRn inhibitor	Efgartigimod	4	84	45.9 (14.4)	63 (75)	3	9.2 (2.6)	16.2 (5.0)	−2.8 (0.5)[b]	−5.3 (0.8)[b]
			Placebo	4	83	48.2 (15.0)	55 (66)	8	8.8 (2.3)	15.5 (4.6)	NA	NA
Howard et al., 2023[8]	3	C5 complement inhibitor	Zilucoplan 0.3 mg/kg	12	86	52.6 (14.6)	52 (60)	28	10.3 (2.5)	18.7 (3.6)	−2.1 (0.6)	−2.9 (0.7)
			Placebo	12	88	53.5 (15.7)	47 (53)	22	10.9 (3.4)	19.4 (4.5)	NA	NA
Meriggioli et al., 2003[30]	Pilot study	IST	Mycophenolate mofetil	35	7	57.7	5 (71)	NA	NA	15.9 (6.0)	NA	−2.6 (2.1)[b]
			Placebo	35	7	51.3	5 (71)	NA	NA	16.7 (3.1)	NA	NA

Continued

Table 1 Summary Statistics of the 27 Included Placebo-Controlled Trials *(continued)*

Study	Phase	Mechanism of action	Treatment	Study length (wk)[a]	n	Age at enrollment, mean (SD)	Female, n (%)	Treatment-related adverse events, n	Baseline MG-ADL score, mean (SD)	Baseline QMG score, mean (SD)	Treatment effect (treatment-placebo), mean (SE)	
											Adjusted MG-ADL score	Adjusted QMG score
Nowak et al., 2022[11]	2	CD20 depletion	Rituximab	52	25	53.2 (17.5)	11 (44)	19	5.8 (3.6)	11.0 (5.1)	0.2 (2.5)	−1.1 (1.3)
			Placebo	52	27	56.8 (17)	12 (44.4)	22	4.0 (3.4)	9.2 (3.9)	NA	NA
Nowak et al., 2024[31]	2	FcRn inhibitor	Batoclimab	6	11	64.5 (18.8)	3 (23.3)	5	8.3 (3.3)	16.4 (2.7)	−3.6 (2.0)[b]	−2.1 (2.2)[b]
			Placebo	6	6	41 (15.4)	4 (66.7)	3	7.3 (3.9)	17.2 (3.2)	NA	NA
Nowak et al., 2025[12]	3	CD19 depletion	Inebilizumab	26	119	47.1 (15.7)	79 (66)	20	9.0 (2.8)	16.7 (4.2)	−1.9 (0.5)	−2.5 (0.7)
			Placebo	26	117	47.9 (15)	65 (56)	22	9.1 (2.8)	17.3 (4.2)	NA	NA
Pasnoor et al., 2016[32]	2	IST	Methotrexate	52	25	NA	6 (24)	NA	4.8 (2.7)	10.5 (4.1)	−1.5 (1.1)	−1.7 (1.6)
			Placebo	52	25	NA	9 (36)	NA	4.1 (3.0)	10.4 (4.2)	NA	NA
Piehl et al., 2022[33]	3	CD20 depletion	Rituximab	24	25	67.4 (13.4)	7 (28)	NA	5.1 (3.2)	9.4 (4.5)	−1.2 (1.0)	−1.1 (1.7)
			Placebo	24	22	58 (18.6)	7 (31.8)	NA	4.5 (2.7)	9.3 (4.2)	NA	NA
Muscle Study Group, 2008[34]	3	IST	Mycophenolate mofetil	12	41	57.1 (18.8)	17 (41.5)	NA	6.9 (3.1)	13.3 (5.6)	−1.0 (0.7)	−0.4 (1.0)
			Placebo	12	39	55.3 (17.7)	16 (41)	NA	7.2 (3.4)	12.5 (4.5)	NA	NA
Tindall et al., 1987[35]	2	IST	Cyclosporine	52	10	64	4 (40)	NA	NA	NA	NA	−9.3 (2.7)[b]
			Placebo	52	10	66.2	3 (30)	NA	NA	NA	NA	NA
Tindall et al., 1993[36]	3	IST	Cyclosporine	42	20	56.05 (19.1)	16 (80)	NA	NA	NA	NA	−4.3 (1.4)[b]
			Placebo	42	19	43.74 (15.9)	11 (57.9)	NA	NA	NA	NA	NA
Vu et al., 2022[9]	3	C5 complement inhibitor	Ravulizumab	26	86	58 (13.8)	44 (51)	29	9.1 (2.6)	14.8 (5.2)	−1.6 (0.5)	−2.0 (0.6)
			Placebo	26	89	53.3 (16.1)	45 (51)	30	8.9 (2.3)	14.5 (5.3)	NA	NA

Continued

Table 1 Summary Statistics of the 27 Included Placebo-Controlled Trials *(continued)*

											Treatment effect (treatment-placebo), mean (SE)	
Study	**Phase**	**Mechanism of action**	**Treatment**	**Study length (wk)[a]**	**n**	**Age at enrollment, mean (SD)**	**Female, n (%)**	**Treatment-related adverse events, n**	**Baseline MG-ADL score, mean (SD)**	**Baseline QMG score, mean (SD)**	**Adjusted MG-ADL score**	**Adjusted QMG score**
Yan et al., 2022[37]	2	FcRn inhibitor	Batoclimab 340 mg	7	10	36.4 (9.8)	8 (80)	NA	7.4 (1.6)	17.4 (3.5)	−2.5 (0.4)[b]	−6.3 (0.6)[b]
			Batoclimab 680 mg	7	11	40.6 (16.8)	9 (81.8)	NA	9.2 (2.3)	18.8 (6.1)	−2.2 (0.4)[b]	−7.1 (0.5)[b]
			Placebo	7	9	40.2 (9.3)	7 (77.8)	NA	8.2 (1.4)	14.9 (5.0)	NA	NA
Yan et al., 2024[38]	3	FcRn inhibitor	Batoclimab 680 mg	6	67	43.8 (13.9)	40 (59.7)	47	8.4 (2.8)	17.9 (4.8)	−1.9 (0.1)[b]	−4.8 (0.1)[b]
			Placebo	6	64	43.7 (13.5)	48 (75)	24	8.5 (2.1)	18.3 (4.9)	NA	NA
Zhou et al., 2017[39]	3	IST	Tacrolimus	24	44	41 (12.8)	28 (63.6)	33	6.2 (3.3)	14.3 (4.3)	−0.9 (0.6)	−1.7 (0.9)
			Placebo	24	38	44 (12.1)	18 (47.4)	29	5.6 (3.1)	13.6 (3.7)	NA	NA

Abbreviations: BAFF = B-cell activating factor; FcRn = neonatal fragment crystallizable receptor; IST = immunosuppressive therapy; IVIg = intravenous immunoglobulin; MG-ADL = Myasthenia Gravis–Activities of Daily Living; QMG = Quantitative Myasthenia Gravis.

[a] Length of study indicates the number of weeks at which the outcome measure (QMG and MG-ADL) treatment effects were reported.

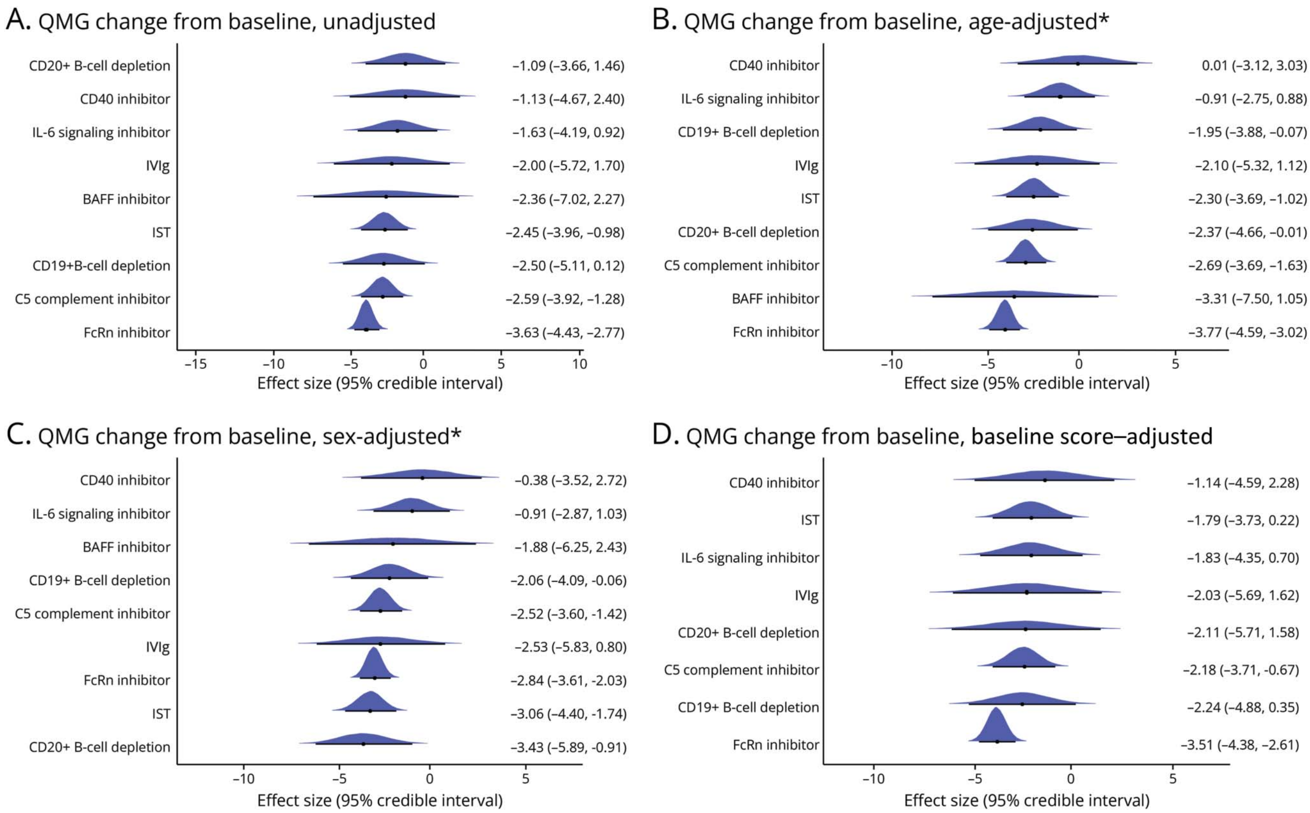
[b] Indicates that adjusted treatment effect was unavailable and the unadjusted treatment effect was used instead.

The unadjusted model estimated that patients treated with FcRn inhibitors would experience −3.63-point (95% CrI −4.43 to −2.77) greater reduction in QMG score than in those treated with placebo and standard of care; −2.59-point (−3.92 to −1.28) reduction for those treated with C5i, and −2.50-point (−5.11 to 0.12) reduction for those treated with CD19$^+$ B-cell depletion. Patients treated with FcRn inhibitors can expect MG-ADL scores to decrease by −1.93 (−2.21 to −1.65) points more than participants treated with placebo and standard of care, −1.91 (−2.97 to −0.84) for those treated with CD19$^+$ B-cell depletion, and −1.84 (−2.48 to −1.19) for those treated with C5i (Figures 2A and 3A). We calculated the relative effects of each treatment's ability to reduce QMG and MG-ADL scores (eFigures 4 and 5). In 100,000 simulations, the probability of FcRn inhibitors having the greatest improvement in QMG and MG-ADL scores compared with all other treatments was approximately 0.9 (eTable 2).

Adjusted Analysis

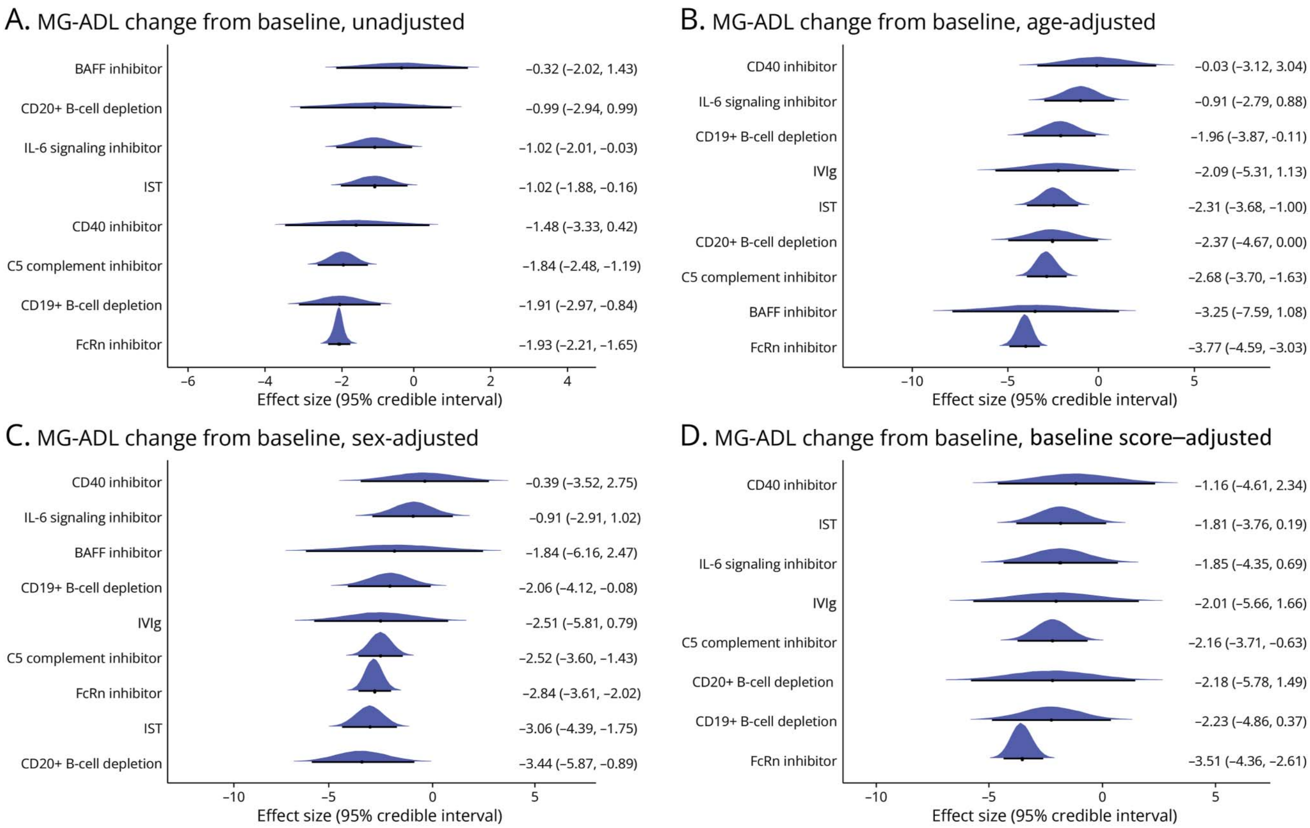
We then analyzed the effect of enrollment age, sex, and baseline QMG and MG-ADL scores on the treatment effects to understand the effects of confounders. We observed that a higher percentage of female patients and greater enrollment age in a study was associated with a greater reduction in QMG scores (sex: β = −3.16; 95% CrI −4.45 to −1.76; age: β = 2.09; 95% CrI 0.78–3.27) (eTable 2). We did not observe a significant association between baseline QMG score and the QMG treatment effect nor did we observe any significant association between covariates and MG-ADL treatment effects. We calculated adjusted treatment effects, which estimated the effect of each treatment under consistent experimental conditions (i.e., if all trials had the same age at enrollment, percentage of female participants, and baseline QMG or MG-ADL scores); these data are presented in Figures 2 and 3. Both FcRn inhibitors (unadjusted −3.63 [−4.43 to −2.77]; age-adjusted −3.77 [−4.59 to −3.02]; sex-adjusted −2.84 [−3.61 to −2.03]), C5i (unadjusted −2.59 [−3.92 to −1.28]; age-adjusted −2.69 [−3.69 to −1.63]; sex-adjusted −2.52 [−3.60 to −1.42]) and CD19$^+$ B-cell depletion (unadjusted −2.50 [−5.11 to 0.12]; age-adjusted −1.95 [−3.88 to −0.07]; sex-adjusted −2.52 [−3.60 to −1.42]) demonstrated marginal changes in the treatment effect after covariate adjustment.

Figure 2 Effect Size, 95% Credible Interval, and Overlaid Posterior Distribution of QMG Change From Baseline (A) Unadjusted, (B) Adjusted for Age at Enrollment, (C) Adjusted for Sex, and (D) Adjusted for Baseline QMG Score



Adjusted effect sizes represent the estimated treatment effect under consistent trial conditions. In this figure, we estimate the treatment effect if all studies had a baseline QMG score of 15.6, an enrollment age of 50.7 years, and a female participant percentage of 57.5% (the averages across all studies with available data). A negative effect size of x is interpreted as follows: the treatment reduces QMG x points more than placebo and standard of care. * Indicates statistically significant β term. BAFF = B-cell activating factor inhibitor; FcRn = neonatal Fc receptor; IL-6 = interleukin-6; IST = immunosuppressive therapy; IVIg = IV immunoglobulin; QMG = Quantitative Myasthenia Gravis.

Figure 3 Effect Size, 95% Credible Interval, and Overlaid Posterior Distribution of MG-ADL Change From Baseline (A) Unadjusted, (B) Adjusted for Age at Enrollment, (C) Adjusted for Sex, and (D) Adjusted for Baseline MG-ADL Score



Adjusted effect sizes represent the estimated treatment effect under consistent trial conditions. In this figure, we estimate the treatment effect if all studies had a baseline MG-ADL score of 7.8, an enrollment age of 49.9 years, and a female participant percentage of 57.5% (the averages across all studies). A negative effect size of x is interpreted as follows: the treatment reduces MG-ADL x points more than placebo and standard of care. * Indicates statistically significant β term; no covariates achieved a statistically significant effect. BAFF = B-cell activating factor inhibitor; FcRn = neonatal Fc receptor; IL-6 = interleukin-6; IST = immunosuppressive therapy; IVIg = IV immunoglobulin; MG-ADL = Myasthenia Gravis–Activities of Daily Living.

The QMG treatment effect for CD20$^+$ B-cell depletion increased after adjusting for enrollment age and sex (unadjusted −0.99 [−2.94 to 0.99]; age adjusted −2.37 [−4.66 to −0.01]; sex adjusted −3.43 [−5.89 to −0.91]) (eTable 2 and Figure 2, B and C).

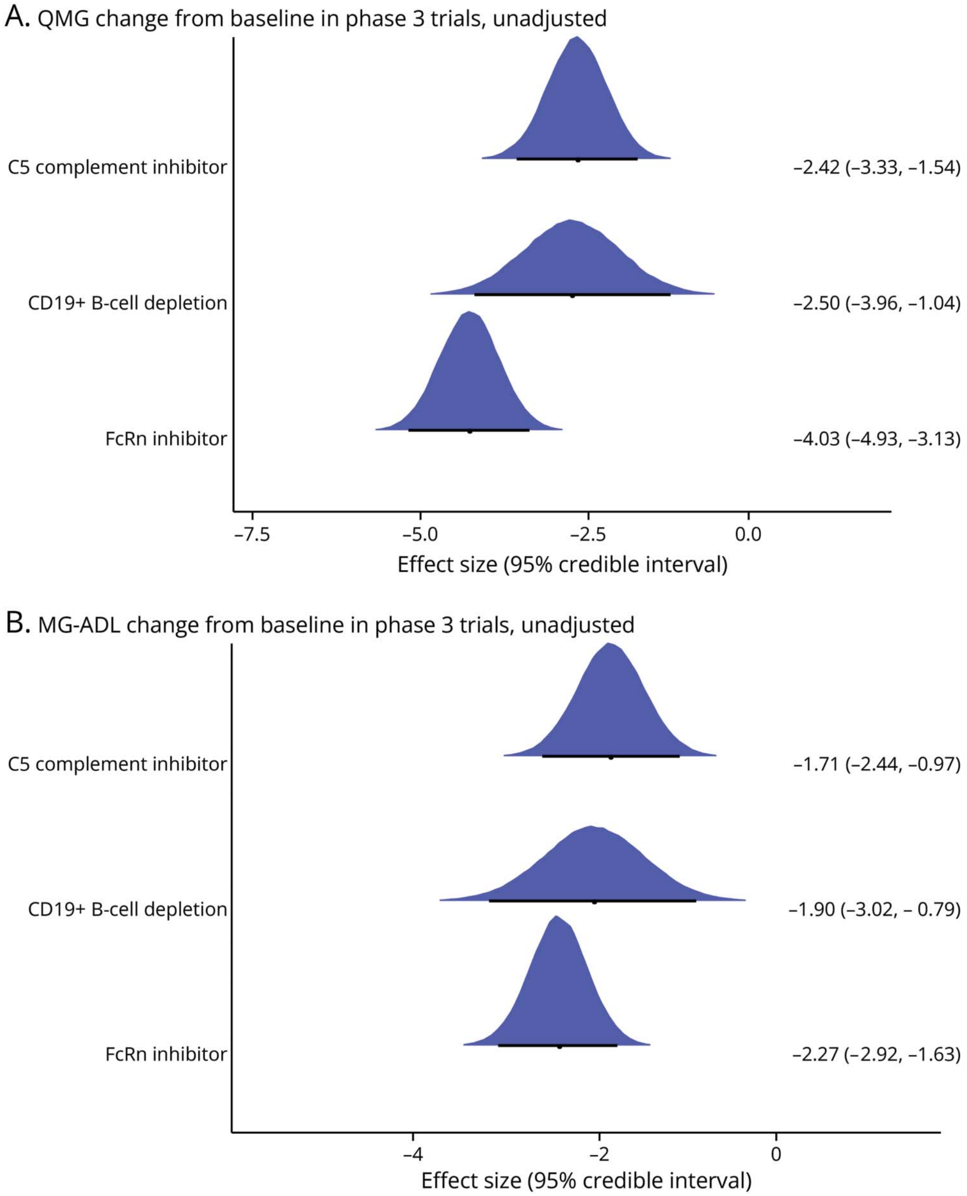
Analysis of Phase 3 Data

The previous analysis used a mixture of phase 2 and 3 trials, but phase 2 trials are typically not powered to assess treatment efficacy. We compared the efficacy of approved targeted therapies based on their performance in phase 3 trials. The FcRn inhibitor treatment class included efgartigimod, nipocalimab, and rozanolixizumab, demonstrating a class-wide QMG treatment effect of −4.03 (−4.93 to −3.13) and an MG-ADL treatment effect of −2.27 (−2.92 to −1.63). CD19$^+$ B-cell depletion only included the phase 3 inebilizumab trial and demonstrated a QMG treatment effect of −2.5 (−3.96 to −1.04) and an MG-ADL treatment effect of −1.9 (−3.02 to −0.79). The C5i class included eculizumab, ravulizumab, and zilucoplan, demonstrating a QMG treatment effect of −2.42 (−3.33 to −1.54) and an MG-ADL treatment effect of −1.71 (−2.44 to −0.97). However, the differences between active treatment arms do not satisfy the definition of a clinically meaningful difference in QMG or MG-ADL scores (Figure 4).

AChR Antibody–Positive Patient Population

We then examined the efficacy of treatments in AChR+ patients. Nineteen studies (n = 798 patients in treatment arms, n = 787 patients in placebo arms) had QMG data, and 15 studies (n = 844 patients in treatment arms, n = 771 patients in placebo arms) had MG-ADL data (eFigures 6 and 7). FcRn inhibitors showed the greatest treatment effect, with treated patients likely, on average, to experience a −3.81-point (−4.89 to −2.71) improvement in QMG scores and a −2.20-point (−2.73 to −1.67) improvement in MG-ADL scores compared with those treated with placebo and standard of care. CD19$^+$ B-cell depletion therapy (BCDT) and C5i had slightly lower QMG treatment effects (CD19$^+$ B-cell depletion −2.50, 95% CrI −4.13 to −0.88; C5i −2.50, 95% CrI −3.36 to −1.63) and MG-ADL treatment effects (CD19$^+$ B-cell depletion −1.79, 95% CrI −2.99 to −0.60; C5i −1.77, 95% CrI −2.42 to −1.13) than FcRn inhibitors but showed comparable efficacy (eFigures 8 and 9). Estimates of the improvement in QMG and MG-ADL scores that AChR+

Figure 4 Efficacy of Approved Targeted Therapies in Phase 3 Trials



C5 complement inhibitor—eculizumab,[10] ravulizumab,[9] and zilucoplan.[8] CD19^{+} B-cell depletion—inebilizumab.[12] FcRn inhibitor—efgartigimod,[19] nipocalimab,[13] and rozanolixizumab.[14] FcRn = neonatal Fc receptor; MG-ADL = Myasthenia Gravis-Activities of Daily Living; QMG = Quantitative Myasthenia Gravis.

patients may experience on one treatment compared with any other are shown in league tables (eFigures 10 and 11).

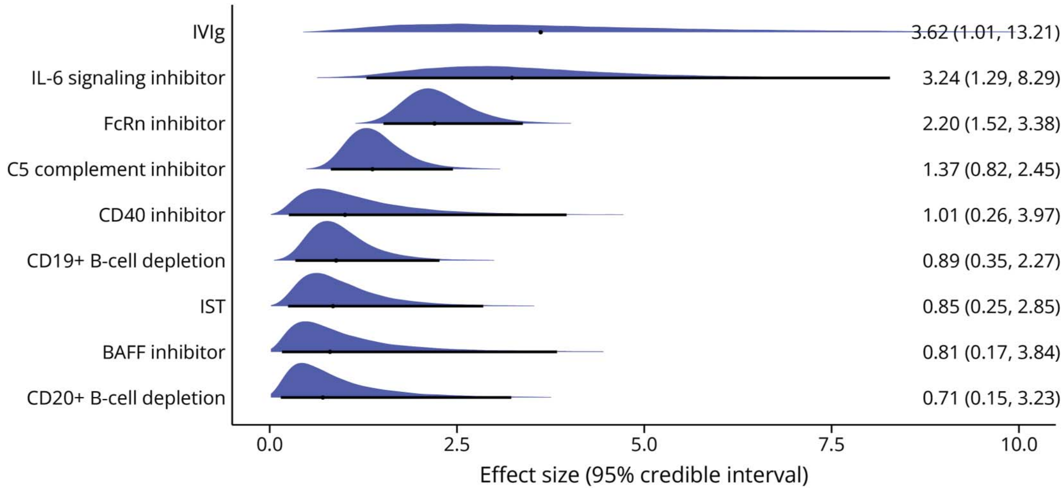
MuSK Antibody–Positive Patients

8Five different treatment regimens from 4 studies reported the treatment difference in MG-ADL change from baseline for MuSK+ patients (n = 59 patients in treatment arms; n = 45 patients in placebo arm) (eFigure 12). Rozanolixizumab 7 mg/kg showed the greatest efficacy in reducing MG-ADL scores compared with placebo and standard of care (−9.42, 95% CrI −15.07 to −3.67) (eFigure 13). We estimated the relative effects of each treatment on MG-ADL reductions in MuSK+ patients and report these findings in a league table (eFigure 14).

Safety

We calculated the OR relative to placebo of TRAEs for studies with the available data (n = 18 studies; n = 1,009 patients in treatment arms; n = 880 patients in placebo arms) (eFigure 15). CD20^{+} B-cell depletion (OR 0.71; 95% CrI 0.15–3.23), the BAFF inhibitor trial (OR 0.81; 95% CrI 0.17–3.84), and IST (OR 0.85; 95% CrI 0.25–2.85) had the lowest odds of TRAE compared with placebo while IVIg (OR 3.62; 95% CrI 1.01–13.21), IL-6 signaling inhibitors (OR 3.24; 95% CrI 1.29–8.29), and FcRn inhibitors (OR 2.20; 95% CrI 1.52–3.38) had higher odds of TRAE compared with placebo (Figure 5). We have calculated the odds of TRAE between treatments and presented these values in a league table (eFigure 16).

Figure 5 Odds Ratio, 95% Credible Interval, and Overlaid Posterior Distribution of Odds of Treatment-Related Adverse Events Compared With Placebo



BAFF = B-cell activating factor inhibitor; FcRn = neonatal Fc receptor; IL-6 = interleukin-6; IST = immunosuppressive therapy; IVIg = IV immunoglobulin.

SUCRA score rankings indicate that CD20$^+$ B-cell depletion has the highest probability (0.735) of having the lowest odds of TRAE followed by IST (0.690) (eTable 3).

Discussion

The landscape of MG treatment has rapidly evolved over the past decade, leaving clinicians to sort through a muddle of therapeutic options. Without a prospective head-to-head study to help clinicians navigate this landscape, we assembled the various randomized trials, both new and old, and compared their efficacies and safety. While existing meta-analyses have reported superiority of FcRn inhibitors, available data do not definitively support this conclusion because meaningful differences in QMG score (3 points) and MG-ADL score (2 points) were not observed between active treatment arms of different classes of agents.[3-6] Considering this nuance and perspective, FcRn inhibitors, C5i, and CD19$^+$ BCDT seem to have comparable efficacy.

Our study also contributes to the current literature through its analysis of the differences between trial populations and their consequent effects on treatment efficacy. Of interest, the QMG treatment effect of CD20$^+$ BCDT improved after adjusting for age and sex. These relationships may be related to the higher frequency of treatment-refractory MG in younger patients and women.[40,41] Refractory patients would fare worse when treated with standard of care in placebo arms, reducing the average improvement seen in placebo arm patients and consequently amplifying the overall study treatment effect (i.e., the difference between treatment and placebo response). This patient population was less prevalent in the 2 rituximab trials which had fewer female patients, older patient populations, and lower baseline QMG and MG-ADL scores compared to the averages across all other studies. For reference, the RINOMAX trial was 29.8% women and had an average age of 63 years at enrollment, average baseline QMG score of 9.35, and an average baseline MG-ADL score of 4.8. The BeatMG study population was 44% women and had an average age of 55.1 years, average baseline QMG score of 10.1, and average baseline MG-ADL score of 4.9.[11,33] The averages across all studies excluding RINOMAX and BeatMG were as follows: 55.6% women, 46.8 years at enrollment, baseline QMG score of 16.1, and baseline MG-ADL score of 8.3.

This claim is further supported by the greater reduction in QMG scores in the placebo arm of the RINOMAX trial compared to the non-rituximab trials (RINOMAX −5.8; non-rituximab trials −1.42).[33] However, this fails to explain the effect, if any, that this vastly different patient population had on the BeatMG study because the placebo arm shows comparable reduction in QMG scores to that observed in other trials (BeatMG −1.7; non-rituximab trials −1.42). This inconsistency suggests that other variables such as frequency of early-onset or late-onset MG and disease duration may better explain the heterogeneity in treatment response. Recent studies have identified pathomechanistic differences between early-onset and late-onset MG.[42,43] Lack of available patient-level data prevented a sensitivity analysis between early-onset and late-onset MG or the effect of disease duration in our study, so for consistency, we elected to use age at enrollment. Future studies should leverage patient-level data to assess the effect of disease duration and onset age on treatment efficacy. Nevertheless, this finding suggests that, under similar experimental conditions to other trials, rituximab may show comparable efficacy to novel targeted therapies; however, without a definitive prospective study, this would be speculative.

Furthermore, differences in mechanisms of action result in temporal variability of treatment response, which add further nuance to any comparison of treatment efficacy. The cyclical pattern of rapid symptomatic improvement followed by relapse that is characteristic of FcRn inhibitors starkly contrasts with the sustained benefit of $CD19^+$ B-cell depletion, which persists to 52 weeks.[12,19] We addressed this in part by conducting a sensitivity analysis using the 52-week randomized controlled period data of $CD19^+$ B-cell depletion in patients with AChR+ and found that the QMG treatment effect improved to −4.3 (−7.05 to −1.54) and the MG-ADL treatment effect improved to −2.79 (−3.99 to −1.61), the highest among all treatment types. Including these data in our analysis is in line with the prespecified analyses for the $CD19^+$ B-cell depletion trial to evaluate possible delayed clinical benefits and durability of response after completion of the forced steroid taper (weeks 4–24).[12] In the interest of navigating the landscape of MG treatment, this finding of comparable treatment effects is encouraging, given the substantially lower treatment burden associated with $CD19^+$ B-cell depletion, which was dosed at baseline, week 2, and week 26, compared with FcRn inhibitors, which require more frequent infusions/injections.[12,13,19] Therefore, the interpretation that a single treatment performed best based solely on clinical outcome measures requires additional contextualization because these measures fail to account for the potential burden of treatment (i.e., cost and frequency of infusions) placed on patients.

The analysis of TRAEs also requires further contextualization. While most treatments seem to show comparable rates of AEs to placebo and to one another, readers may comment on the greater odds of TRAEs in patients treated with IVIg shown in Figure 5. These data came from a single IVIg trial and should therefore be interpreted with caution because IVIg is generally believed to be well tolerated.[24,44] Why patients in this particular trial experienced such a high rate of TRAEs remains unanswered. Similar concerns arise when examining the safety data in patients treated with FcRn inhibitors. The data across trials seem to be driven by the phase 3 trial of batoclimab, which reported a higher rate of TRAEs in the treatment arm compared with placebo.[38] This is a limitation inherent to meta-analyses because not all trials reported TRAEs, preventing a more thorough comparison of treatment safety.

Despite our efforts to conduct a valid comparison of MG treatments, our analysis still has several limitations. Notably absent from this meta-analysis are the azathioprine and thymectomy randomized controlled trials, 2 widely used interventions in the treatment of MG, which were excluded because they reported median, rather than mean, change from baseline in QMG and MG-ADL scores.[45,46] One 2008 trial of mycophenolate mofetil was excluded because it did not report change from baseline in the outcome measures.[47] In addition, a more thorough comparison of the novel therapies with IVIg, using data from additional IVIg trials, was not feasible, given differences in trial outcomes.[48-50] Second, some studies did not report adjusted least squares mean treatment differences. In these cases, we used the unadjusted treatment difference. Third, Bayesian hierarchical models are computationally expensive, thus limiting the meta-regression complexity in the packages we used to simple linear regression. We would like to emphasize that the covariate-adjusted treatment effects are model-based predictions rather than subgroup analyses and should not be misinterpreted as Class I evidence sourced from randomized trials. Finally, some features of individual trials remain unaddressed. Some studies, such as the $CD19^+$ B-cell depletion trial, used scheduled corticosteroid tapering regimens while others reduced corticosteroid dosages at the investigator's discretion. In addition, 2 trials of mycophenolate mofetil, 2 trials of cyclosporine, 1 trial of tacrolimus, and 2 trials of batoclimab were not multinational, which may reduce their generalizability to the broader patient population.[30,34-39]

Our study shows that FcRn inhibitors, $CD19^+$ B-cell depletion, and C5i have comparable efficacies. However, our covariate-adjusted analysis is preliminary, and we hope that it incites more robust analyses and future discussion into the influence of trial design and populations on treatment efficacy. When choosing a therapeutic intervention, clinicians must consider individual patients in the context of their age, sex, and disease severity, as well as the ease of therapeutic administration and cost of treatment. That said, only a prospective study can thoroughly explicate the causes of subtle variations in treatment response and allow clinicians to determine optimal treatment strategies for a given patient. Thus, a longitudinal, prospective, comparative effectiveness study is warranted to allow the collection, not only of traditional trial end points (i.e., MG-ADL and QMG scores) but also of data on disease course, burden of disease/treatment, and long-term impact of therapy on outcomes.

Author Contributions

N.P. McLaren: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. M. Rosati: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. W. Zhong: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. S. Hussain: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. M.C. Funaro: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. R.J. Nowak: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. B. Roy: drafting/revision of the manuscript for content, including medical writing for content; study concept or design.

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